



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

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Section: [R]

Group No: 05

Project Report On

Delivery Management System

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CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				Total =
Requirement fulfillment	Properly demonstrate a real-life scenario-based project with proper functional requirement identification for the Object-Oriented Programming project development activities.				
Validation	Ensuring the ability of students' proper demonstration on validation forms in their system in terms of dealing with the data.				
Verification	Identifying if the students can verify the system data along with proper functional requirements in terms of data flow.				

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CHAPTER 01: INTRODUCTION

The **Delivery Management System** is a database-driven application designed to manage shipment orders from sellers to customers, ensuring accurate payment verification, carrier assignment, tracking, and delivery. It supports four main user roles — **Admin, Employee, Carrier, and Client** — with clearly defined responsibilities and permissions. This system successfully integrates all the essential components of a modern logistics platform. By combining **Client, User, Shipment, Route, Payment, Carrier, and Employee** modules into a normalized SQL Server database, the system ensures data integrity, scalability, and operational efficiency.

The workflow is designed to be linear and predictable: sellers create shipments, employees verify payments and assign carriers, carriers update tracking, and customers securely view their deliveries. Role-based permissions guarantee that each user — Admin, Employee, Carrier, or Client — operates within clearly defined boundaries, ensuring both security and accountability.

Through automation of critical processes such as route assignment, payment validation, and estimated delivery calculation, the system reduces manual errors and improves transparency for all stakeholders. Sellers gain visibility over all their orders, customers can track only their own shipments, employees manage the operational flow, and admins maintain full oversight.

In conclusion, the Delivery Management System provides a robust, reliable, and user-friendly foundation for logistics providers. It not only streamlines shipment handling but also enhances trust by giving each role the right level of access and control.

CHAPTER 02: USER STORY

Client's User Stories

- Can log in securely to access the client dashboard.

➤ **Seller's User Stories**

- Can create a shipment order by entering goods type, weight, volume, pickup/drop-off addresses, and customer details.
- System records the shipment in the Shipment table.
- System automatically assigns a RouteID for each unique pickup/drop-off combination.
- Can view all shipment orders they have ever created.

➤ **Customer's User Stories**

- Existing Member:
 - Can log in securely to view shipment details.
 - Can track only the shipments linked to their ClientID.
- Not Yet a Member:
 - Can verify shipment using name and email.
 - System checks details against the Shipment table.
 - If not registered, redirected to a SignUp form.
 - Can create an account by providing contact number, password, date of birth, gender, and address.
 - System stores new information in the Client table.
- New User's User Stories:
 - Can sign up by filling out a registration form.
 - Can create an account and log in to the system.
 - System inserts details into the Client table upon pressing the Create button.

Employee's User Stories

- Can log in securely to access the employee dashboard.
- Can verify whether a shipment is fully paid or unpaid.
 - System checks and updates the Payment table.
- Can assign shipment orders to carriers only after payment is confirmed.
 - System updates the Shipment table with the assigned CarrierID.
- Can update the tracking status of shipments.
 - System automatically assigns an Estimated Delivery Date = Current Date + 3 days once status changes from Pending.
- Can add or delete carriers and shipments to manage operational logistics.

Carrier's User Stories

- Can log in securely to access the carrier dashboard.
- Can update the tracking status of shipments.
- System records updates so customers and employees can monitor delivery progress.

Admin's User Stories

- Can log in securely to access the admin dashboard.
- Can oversee and modify shipment, payment, and tracking records.
 - Ensures data accuracy and resolves disputes.
 - Can update all fields in the Payment table.
- Can add or remove employees, carriers, and shipments to maintain organizational control.
- Cannot delete client accounts; only clients can delete their own accounts.

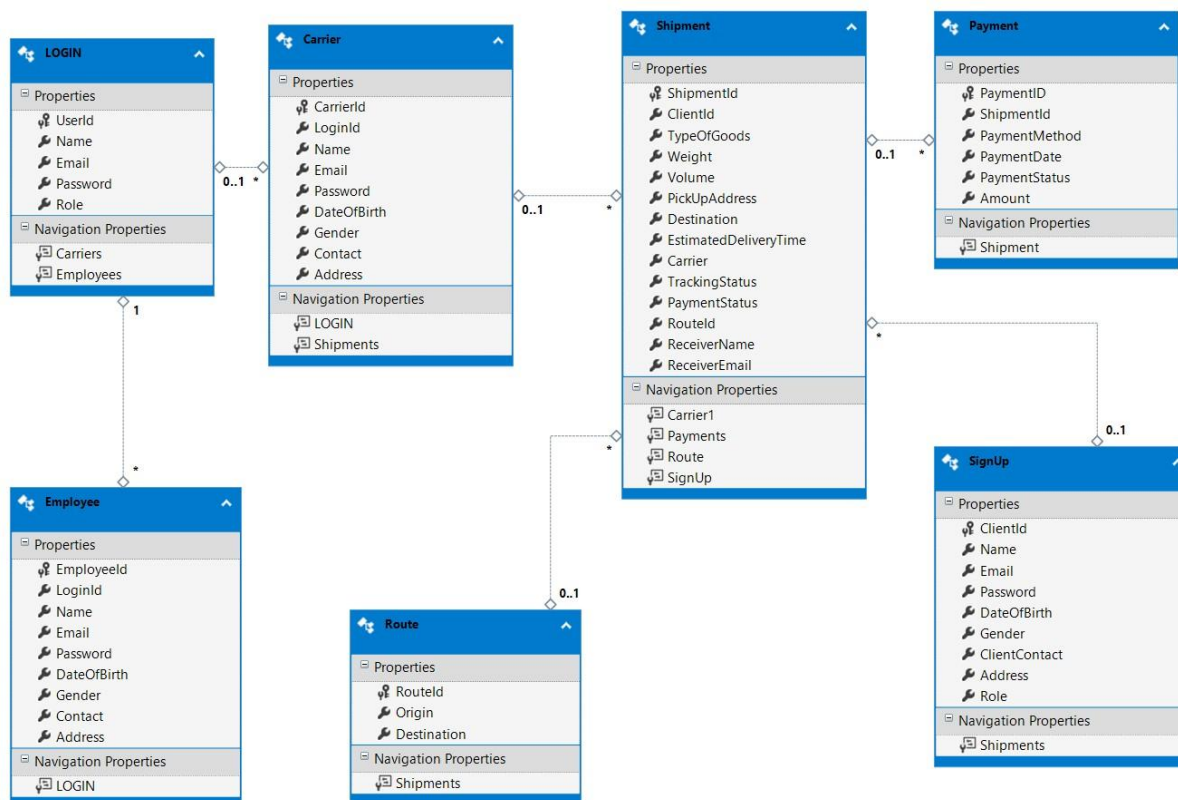
❖ Schema summary:

Table	Key Fields	Relationships / Notes
SignUp	ClientID (PK), Name, Email, ContactNo, Password, DOB, Gender, Address, Role (Client)	Stores both Sellers and Customers . Seller creates shipments; Customer receives them.
Login	UserID (PK), Name, Email, Role (Admin/Employee/Carrier), Password	Stores login credentials for Admins, Employees, and Carriers (separate from Clients).
Shipment	ShipmentID (PK), ClientID (FK), RouteID (FK), TypeOfGoods, Volume, Weight, CarrierID (FK), PaymentStatus, TrackingStatus, PickupAddress, Destination, EstimatedDeliveryTime, ReceiverEmail, ReceiverName	Created by Seller. Linked to Client, Route, Carrier. Payment status defaults to Pending . Estimated delivery auto-set = Today + 3 days when payment status is changed from Pending to Paid/Unpaid.
Route	RouteID (PK), Origin, Destination	Auto-assigned for each unique pickup/drop-off combination set by seller.
Payment	PaymentID (PK), ShipmentID (FK), PaymentMethod, PaymentDate, Amount, PaymentStatus	Editable by Admin/Employee. Shipment must be Paid/Unpaid before carrier assignment.
Carrier	CarrierID (PK), Name, Email, ContactNo, Password, DOB, Gender, Address, UserID (FK)	Added by Admin/Employee. Assigned to shipments. Updates Tracking Status (In Transit, Shipped, Delivered).
Employee	CarrierID (PK), Name, Email, ContactNo, Password, DOB, Gender, Address, UserID (FK)	Added by Admin. Updates Payment Status and appoints Carrier to shipments.

❖ Role permissions at a glance:

Role	Can Create	Can View	Can Update	Can Delete
Admin	Employees, Carriers, Shipments, Routes	All records	Payment, Tracking, Carrier & Employee info, Route, Shipment assignment	Employees, Carriers, Shipments (not Clients)
Employee	Carriers, Shipments	All shipments	Payment (all fields), Assign Carrier, Route, and own profile info	Carriers, Shipments (not Clients)
Carrier	–	Assigned shipments	Tracking status and own profile info	–
Seller (Client)	Shipments	All their shipments	Own profile info	Own account
Customer (Client)	–	Only shipments tied to their ClientID	Own profile info	Own account

CHAPTER 03 (a): ER DIAGRAM



CHAPTER 03 (b): SQL Queries

SELECT QUERY:

1. SELECT * FROM Signup WHERE Email = {email};
2. SELECT Password FROM Signup WHERE Email = '{Email}';
3. select * from Payments;
4. SELECT PaymentStatus FROM Shipment WHERE ShipmentId = txtId.Text;
5. SELECT ClientId FROM SignUp WHERE Email='{staticWelcome.email}';
6. select distinct Origin from Route;
7. SELECT Role, Name FROM Signup WHERE Email = '{email}' AND Password = '{password}';
8. \$"SELECT * FROM Signup WHERE Email = '{email}';
9. "SELECT * FROM employee WHERE EmployeeId =" + Id;
10. \$"SELECT * FROM Signup WHERE Email = '{email}';

UPDATE QUERY:

1. UPDATE SignUp SET "Password = " + password + ", " + "DateOfBirth = " + dob + ", " + "Gender = " + gender + ", " + "ClientContact = " + contact + ", " + "Address = " + address + " " + "WHERE ClientId = " + ClientId;
2. UPDATE Shipment SET TrackingStatus=" + cmbTrackingStatus.SelectedItem + ",TypeOfGoods=" + txtGoods.Text + ",EstimatedDeliveryTime=" + dateTimePicker1.Value + ",Carrier=" + cmbCarrier.SelectedValue + " WHERE ShipmentId=" + txtId.Text;
3. \$"UPDATE Carrier SET Name='{name}', Password='{password}', DateOfBirth='{dob:yyyy-MM-dd}', Gender='{gender}', Contact='{contact}', Address='{address}' WHERE LoginId = {Id}";

INSERT QUERY:

1. INSERT INTO Signup (Name, Email, Password,DateOfBirth,Gender,ClientContact,Address, Role) " + \$"VALUES ('{name}', '{email}', '{password}', '{dob:yyyy-MM-dd}', '{gender}' , '{contact}', '{address}','Client')";
2. INSERT INTO SignUp (Name, Email, Role) "+ \$"VALUES ('{RName}', '{REmail}', 'Client')";
3. INSERT INTO Route (Origin,Destination) " VALUES ('{Origin}','{Destination}')";
4. INSERT INTO Employee(LoginId,Name,Email,Password,DateOfBirth,Gender,Contact,Address)" + \$"Values('{Id}','{Name}', '{email}', '{Password}', '{dob}','{gender}','{Contact}','{Address}');"

DELETE QUERY:

1. delete from Shipment where ShipmentId=" + txtId.Text;
2. \$"DELETE FROM Payments WHERE PaymentID={txtPaymentId.Text}";
3. delete from LOGIN where UserId= + Id;
4. \$"DELETE FROM Signup WHERE Email = '{email}';

CHAPTER 04: Screenshots

1. LOGIN PAGE VIEW



WELCOME TO
SMS LOGISTICS &
TRANSPORTS

SMS LOGISTIC AND
TRANSPORT COMPANY

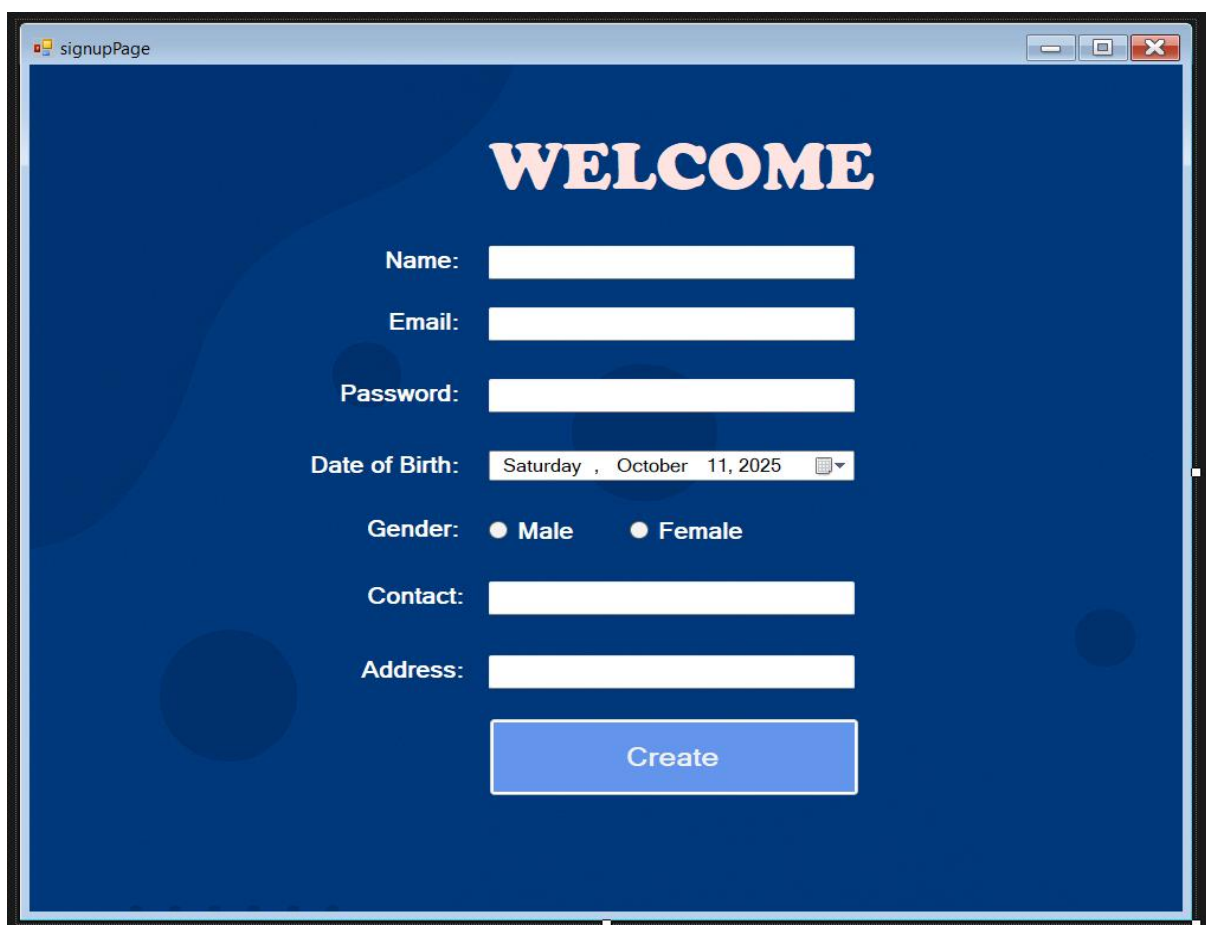
EMAIL

PASSWORD

LOGIN

SIGN UP

2. SIGNUP PAGE VIEW



WELCOME

Name:

Email:

Password:

Date of Birth:

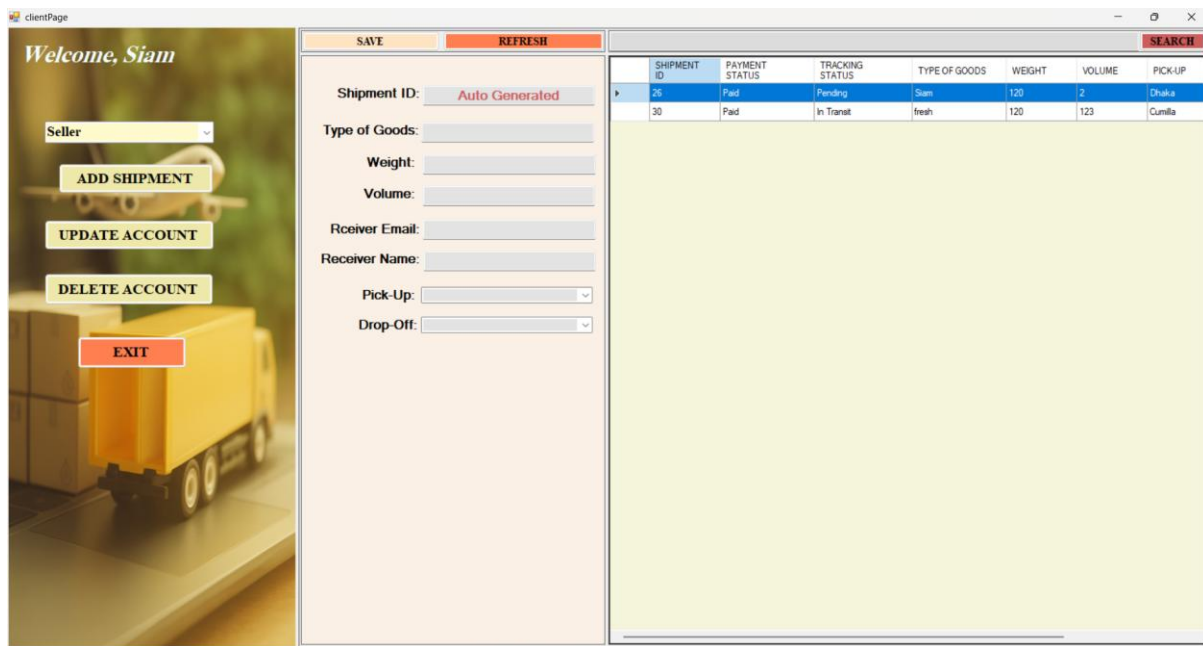
Gender: ☒ Male ☐ Female

Contact:

Address:

Create

3. CLIENT PAGE (SELLER VIEW)



The screenshot shows the 'clientPage' window for a seller. On the left is a sidebar with a 'Welcome, Siam' header, a 'Seller' dropdown menu, and buttons for 'ADD SHIPMENT', 'UPDATE ACCOUNT', 'DELETE ACCOUNT', and 'EXIT'. The main area has a 'SAVE' button and a 'REFRESH' button. Below these are input fields for 'Shipment ID' (showing 'Auto Generated'), 'Type of Goods', 'Weight', 'Volume', 'Receiver Email', 'Receiver Name', 'Pick-Up', and 'Drop-Off'. On the right is a table with columns: SHIPMENT ID, PAYMENT STATUS, TRACKING STATUS, TYPE OF GOODS, WEIGHT, VOLUME, and PICK-UP. The table contains two rows of data.

SHIPMENT ID	PAYMENT STATUS	TRACKING STATUS	TYPE OF GOODS	WEIGHT	VOLUME	PICK-UP
26	Paid	Pending	Siam	120	2	Dhaka
30	Paid	In Transit	fresh	120	123	Cunilla

4. CLIENT PAGE (CUSTOMER VIEW)



The screenshot shows the 'clientPage' window for a customer. The sidebar is identical to the seller view, but the dropdown menu is set to 'Customer' and the buttons are 'SHOW DETAILS', 'UPDATE ACCOUNT', 'DELETE ACCOUNT', and 'EXIT'. The main area has a 'REFRESH' button and a 'SEARCH' button. Below these is a table with columns: SHIPMENT ID, PRODUCT TYPE, TRACKING STATUS, WEIGHT, VOLUME, PICK-UP, DROP-OFF, EST. DELIVERY TIME, and PAYMENT STATUS. The table is currently empty.

SHIPMENT ID	PRODUCT TYPE	TRACKING STATUS	WEIGHT	VOLUME	PICK-UP	DROP-OFF	EST. DELIVERY TIME	PAYMENT STATUS
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5. CLIENT PAGE (UPDATE ACCOUNT VIEW)

Welcome, Siam

Customer
SHOW DETAILS
UPDATE ACCOUNT
DELETE ACCOUNT
EXIT

CLIENT INFORMATION

Client ID: 2
Name: Siam
Email: siam1234@gmail.com
Password: 12345
Date of Birth: Wednesday, June 16, 2004
Gender: ☒ Male ☐ Female
Contact Number: 01754548547
Address: Kuril
UPDATE

6. EMPLOYEE PAGE (SHIPMENT VIEW)

Employee
Welcome, Mahabubur Rahman

SHIPMENT
ADD CARRIER
ADD ROUTE
MY INFO
EXIT

PAYMENT
CONFIRM
LOAD
DELETE
SEARCH

Shipment ID: Auto Generated
Client ID:
Route ID:
Types Of Goods:
Weight:
Volume:
Pick-Up:
Drop-Off:
Est. Del. Date: Wednesday, October 15
Carrier:
Tracking Status:

Shipmentid	Clientid	TypeOfGoods	Weight	Volume	Pick-Up-Address	Destination	EstimatedDate
23	18	Fish	20	3	Rangpur	Dhaka	10/14/2025
24	19	Car	300	30	Rangpur	Dhaka	10/14/2025
25	19	Fish	20	3	Dhaka	Cumilla	10/13/2025
26	2	Siam	120	2	Dhaka	Rangpur	10/14/2025
27	19	Table	20	2	Cumilla	Dhaka	10/14/2025
28	19	Chair	0.5	2	Dhaka	Cumilla	
29	19	Apple	10	4	Dhaka	Cumilla	
30	2	fresh	120	123	Cumilla	Dhaka	10/14/2025
31	19	Table	12	3	Cumilla	Dhaka	
32	19	Water	100	10	Rangpur	Dhaka	

7. PAYMENT PAGE VIEW

Payment

SAVE REFRESH DELETE

SEARCH

Payment ID: Auto Generated

Shipment ID:

Amount:

Payment Method:

Payment Date: Sunday , October 12, 2025

Payment Status:

PaymentID	ShipmentId	PaymentMethod	PaymentDate	PaymentStatus	Amount
15	24	Cash	10/10/2025	Paid	1000
17	23	Cash	10/10/2025	Paid	100
20	25	Bkash	10/10/2025	Paid	100
21	26	Cash	10/11/2025	Paid	100
22	27	Nagad	10/11/2025	Paid	1200
23	30	Bkash	10/11/2025	Paid	1233

8. ADD CARRIER PAGE VIEW

CarrierAdd

SAVE REFRESH DELETE

SEARCH

ID: Auto Generated

Name:

Email:

Password:

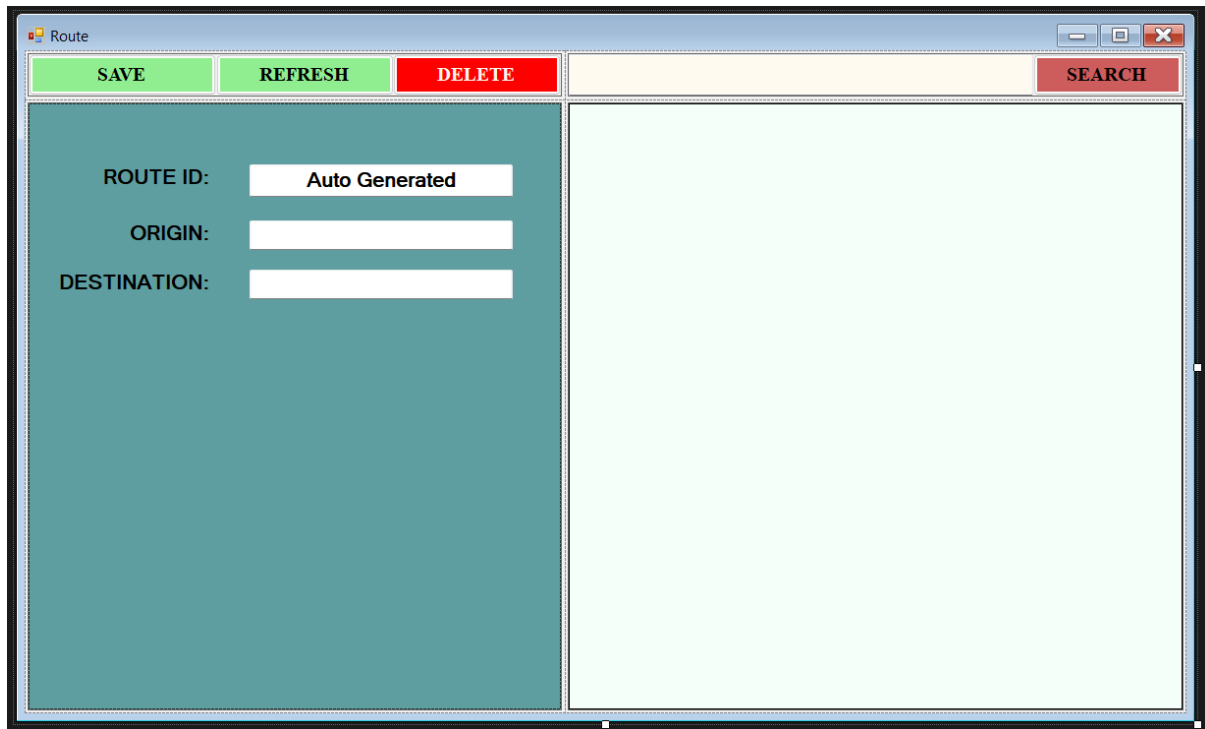
Date of Birth: Saturday , October 11, 2025

Gender: ☐ Male ☐ Female

Contact:

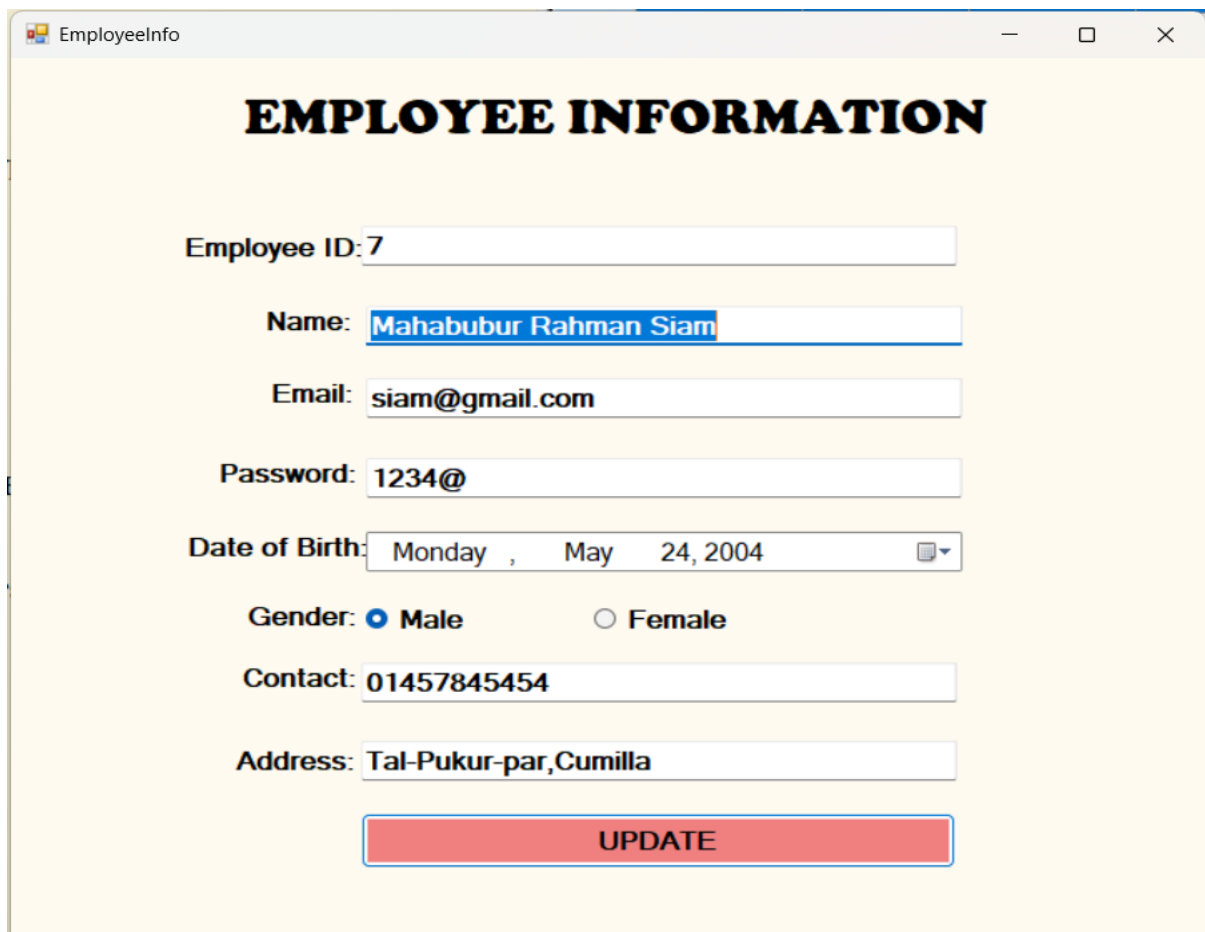
Address:

9. ROUTE PAGE VIEW



The screenshot shows a web application window titled "Route". At the top, there are four buttons: "SAVE" (green), "REFRESH" (green), "DELETE" (red), and "SEARCH" (red). Below these buttons, the window is divided into two main sections. The left section has a teal background and contains three input fields: "ROUTE ID:" with the value "Auto Generated", "ORIGIN:" (empty), and "DESTINATION:" (empty). The right section has a light green background and is currently empty.

10. EMPLOYEE INFORMATION VIEW



The screenshot shows a web application window titled "EmployeeInfo". The main heading is "EMPLOYEE INFORMATION". Below the heading, there are several input fields and a form: "Employee ID:" with the value "7", "Name:" with the value "Mahabubur Rahman Siam", "Email:" with the value "siam@gmail.com", "Password:" with the value "1234@", "Date of Birth:" with a date picker showing "Monday , May 24, 2004", "Gender:" with radio buttons for "Male" (selected) and "Female", "Contact:" with the value "01457845454", and "Address:" with the value "Tal-Pukur-par,Cumilla". At the bottom, there is a red button labeled "UPDATE".

11. CARRIER PAGE VIEW (INFO & UPDATE, APPOINTMENT)

The screenshot shows the 'carrierPage' interface. On the left is a sidebar with a background image of a warehouse. It contains a welcome message 'Welcome, JOHIR ISLAM' and navigation buttons: 'INFO', 'APPOINTMENT', 'UPDATE', and 'EXIT'. The 'INFO' button is selected. The main area displays a table with shipment data and a 'CarrierInfo' modal window.

ShipmentId	ClientId	TypeOfGoods	Weight	Volume	Pick Up Address	Destination	EstimatedDeliveryT	Carrier	TrackingStatus	PaymentStatus
30	2	fresh	120	123	Cumilla	Dhaka	10/14/2025 5:20...	7	In Transit	Paid

CARRIER INFORMATION

Carrier ID: 7

Name: JOHIR ISLAM

Email: h@gmail.com

Password: 12345

Date of Birth: Monday, January 2, 1989

Gender: ☒ Male ☐ Female

Contact Number: 01754854548

Address: Kuril,Dhaka

UPDATE

12. ADMIN PAGE VIEW

The screenshot shows the 'admin' interface. On the left is a sidebar with a background image of a warehouse. It contains a welcome message 'WELCOME, Sumaiya' and navigation buttons: 'ADD EMPLOYEE', 'ADD SHIPMENT', 'ADD CARRIER', 'ADD ROUTE', and 'EXIT'. The 'ADD EMPLOYEE' button is selected. The main area displays a table with employee data and a form for adding a new employee.

EmployeeId	LogInId	Name	Email	Password	DateOfBirth	Gender	Contact
4	37	Sultana Razia M...	razia@gmail.com	1234	5/6/1995	Female	01754854...
5	38	Samaher	samaher@gmail...	1234	10/19/2012	Female	01795658...
7	43	Mahabubur Rah...	siam@gmail.com	1234@	5/24/2004 6:57 ...	Male	01457845...

WELCOME, Sumaiya

ADD EMPLOYEE

Employee ID: Auto Generated

Name:

Email:

Password:

Date of Birth: Sunday, October 12, 21

Gender: ☐ Male ☐ Female

Contact:

Address:

REFRESH **SAVE** **DELETE** **SEARCH**